

To do: Receive a grade

Due: Monday, 13 July 2026, 11:55 PM

This assignment has two parts. All components must be submitted by the end of Unit 11. This assessment accounts for 40% of your final module mark. Each student must submit an individual presentation (20 minutes).

Assignment Topic: Comparative Intelligence and Ethics in Machine Learning Systems

The assignment places particular emphasis on:

- Trade-offs between classical ML and DL approaches.
- Model evaluation and validation.
- Explainability and interpretability of models.
- Ethical implications and bias.
- Considerations for real-world deployment

This assessment requires both technical implementation and critical reflection, communicated through a 20-minute individual presentation and supporting materials.

Dataset Selection

Students must select a real-world dataset from a reputable public source such as Kaggle, the UCI Machine Learning Repository, or AWS Open Data.

You must choose one of the following application tracks:

Track A: Computer Vision (CV)

Tasks may include object recognition, anomaly detection, or image classification.

Example Datasets:

- Chest X-Ray Images (<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>).
- CIFAR-10 (<https://www.cs.toronto.edu/~kriz/cifar.html>).
- Fashion-MNIST (https://www.tensorflow.org/datasets/catalog/fashion_mnist)

Track B: Recommender Systems (RS)

Tasks may include rating prediction or item ranking.

Example Datasets:

- MovieLens (100K or 1M; <https://grouplens.org/datasets/movielens/100k/>).
- Spotify Selection Dataset (<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>).

Students are responsible for justifying their dataset choice and describing any preprocessing steps applied.

Modelling Requirements

You must implement and compare TWO distinct models, one from traditional ML and one for DL, using the same dataset:

- **Communicate Technical Findings Clearly and Effectively**

Students must justify all key design decisions, including architecture, hyperparameters, and training strate

Chat to us!

- **Evaluation, Validation and Explainability**

Models must be evaluated and compared using appropriate methodologies:



- Performance Metrics
- Computer Vision: Accuracy, Precision, Recall, F1-Score.
- Recommender Systems: RMSE, NDCG (or equivalent).

■ Validation Strategies

Apply k-fold cross-validation or an equivalent robust validation approach.

Explain why validation is critical for assessing generalisation and avoiding overfitting.

■ Explainable AI (XAI)

Students must apply at least one explainability technique, such as:

- SHAP or LIME (model-agnostic).
 - Grad-CAM (for CNN-based vision models).
- Explainability results should be used to interpret model behaviour and support comparative analysis.

■ Bias and Diagnostic Analysis

Students must investigate potential performance disparities across subgroups (e.g. lighting conditions in CV datasets or low-activity users in recommender systems).

■ Ethical Considerations

Your analysis must address ethical implications relevant to your chosen application, including:

- Data Privacy: Data provenance, consent, and presence of sensitive or personally identifiable information (PII).
- Algorithmic Fairness: Real-world consequences of false positives and false negatives in your application context.
- Bias and Harm: Potential risks arising from biased data or model decisions.
- Ethical reflection should be grounded in the specific use case rather than treated abstractly. It should also include justified recommendation of which model is most suitable for real-world deployment, linked to the next section.

■ You are not required to implement deployment, but you must discuss how your selected model could be deployed in a real-world setting. This may include:

- Cloud ML platforms (e.g. AWS SageMaker, Google Vertex AI, Azure ML).
- Use of APIs for model inference.
- Scalability, monitoring, and retraining considerations.
- Tools and Libraries

Students must justify their choice of tools. Recommended options include:

- Deep Learning: PyTorch or TensorFlow/Keras.
- Classical ML: Scikit-learn.
- Explainability: SHAP, LIME, InterpretML

Deliverables

1. Code: Mandatory.

A Jupyter Notebook (.ipynb) containing clean, well-documented code.

The notebook should include data preprocessing, model training, evaluation, and explainability analysis,

2. Presentation: Mandatory.

A 20-minute individual recorded presentation with slides and a transcript.

The presentation must clearly cover and include these sections:

- Dataset selection and preprocessing.
- Model design (classical ML vs DL).
- Evaluation metrics and validation strategy.
- Explainability findings.
- Ethical and bias considerations.
- Deployment considerations.



Chat to us!

Slides should prioritise diagrams, visualisations, and results rather than dense text.

Clarity of spoken explanation is weighted more heavily than slide content.

Presentation Format and Submission Guidelines

For this assignment, you are required to record a 20-minute presentation with a transcript. The presentation must include:

- Clear, structured explanations of model development.
- Visual aids (diagrams, tables, results, performance graphs).
- Live or pre-recorded demonstration of the model.
- Comparative insights with other ML approaches.

Important Submission Notes

- Your slides should not be overly text-heavy. Instead, focus on illustrations, statistics, and key takeaways.
- Oral presentation is weighted higher than slides, so clarity and engagement matter.
- It is highly recommended to use a headset with a microphone for clear audio.
- The transcript should be submitted alongside the presentation.
- The maximum file size for your presentation should be below 500MB. Zipped files must not be submitted. If your submission includes multiple components, each file must be uploaded individually.

Word Count Guidance

- Oral presentation length: 1500 words for a typical speaking speed.
- Slide deck: No fixed word count, but should be visually engaging.

Learning Outcomes

- Apply and critically appraise machine learning techniques to real-world problems, particularly where technical risk and uncertainty is involved.
- Critically compare classical ML and Deep Learning approaches.
- Evaluate and validate models using appropriate metrics.
- Interpret model behaviour using explainable AI methods.
- Reflect on ethical and deployment considerations in AI systems.
- Communicate technical findings clearly and effectively

Turnitin Originality Check

Before submitting your assignment, it is important to check the originality of your work by submitting your assignment to [Turnitin](#).

By submitting your assignment to this tool you will receive an originality report which can be used to check that you have not included other authors work without correct citation. It is important to note that submitting your work to the Turnitin Originality Check tool does not count as a submission of your final work. You must still submit your assignment below.

Academic Integrity and Plagiarism

We take academic integrity very seriously. Academic integrity means acting with fairness and honesty, giving credit to others where you are referring to their ideas or research and respecting the work of others. Plagiarism is defined as: 'Using or copying the work of others (whether written, printed or in any other form) without proper acknowledgement'. Before you finalise your assignment take time to check that all your statements are backed up with supporting evidence, that all sources you use - whether referring to their ideas, quoting directly or paraphrasing - are correctly referenced in the text. Correct use of referencing acknowledges the academic whose work has informed yours, enables the reader to find the sources you have used and demonstrates your ability to find and analyse relevant information.

Failure to properly acknowledge the work of others is an academic offence and may result in your work incurring a penalty or, in the most serious cases, you being removed from the course for academic dishonesty.

If you are unsure about referencing or plagiarism there are useful resources available in the Study Skills Hub with [Chat to us!](#) in the menu on the left hand side. If you are still experiencing difficulties with academic integrity then you can [contact the Study Skills team](#) for individualised support.

Submission Instructions

- Submit your saved document below before the end of Unit 11.
- After the deadline, the submission page will be locked.
- Use Harvard referencing for all citations and references.
- If you need to apply for Late Submission, please complete the [late submission of coursework form](#).

Submission status

Attempt number	This is attempt 1.
Submission status	Submitted for grading
Grading status	Graded
Time remaining	Assignment was submitted 2 days 2 hours early
Last modified	Saturday, 11 July 2026, 9:51 PM
File submissions	<u>Final Submission Checklist.docx</u> 11 July 2026, 9:51 PM  Turnitin ID: 2996476265 <u>FINAL_SUBMISSION_CHECKLIST.md</u> 11 July 2026, 9:51 PM  This file has not been submitted to Turnitin because the file format is not supported <u>presentation.pptx</u> 11 July 2026, 9:51 PM  Turnitin ID: 2996476243
Submission comments	> <u>Comments (1)</u>

Feedback

Feedback comments	+ The submission demonstrates a critical structural failure regarding compliance with the mandatory assignment requirements. While the ...
-------------------	--



Chat to us!

You are logged in as Sarra Fadhl Mohamed Binhadda Alsuwaidi (Log out)

[Policies](#)

Powered by Moodle

[Site Accessibility Statement](#)

[Privacy Policy](#)

© 2026 University of Essex Online. All rights reserved.



Chat to us!